

GAP-Net: Geometric-Aware Prompt Calibration Network for Aerial-Ground Person Re-Identification

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Abstract

In the process of aerial ground person re identification, the biggest problem is that there is a big gap between drones and fixed cameras, which will lead to two problems. One is that the human body is deformed, the other is that the human body changes, resulting in semantic ambiguity. At present, many scholars have focused on the research of these two problems, but the two problems are actually related, so the approach of solving them separately is not feasible.

In this study, a new framework is constructed, which can integrate geometric prior estimation and adaptive semantic prompt learning in a natural way, and the framework is named GAP-Net. The whole method is divided into three parts. The first part is the geometric prior extraction module, which can encode the deformation field into geometric embedding, but will not interfere with the original feature map. The second part is the geometric awareness calibration module, which injects geometric prior into the calibration process, and the prompt has geometric awareness and semantic invariance. The third part is the local feature refinement module, which can mine local features through bidirectional attention mechanism with the help of geometric awareness prompt.

In the experiment, many data sets are used, and the evaluation indicators are set in different ways. It is found that the performance of GAP-Net is relatively ideal. Ablation experiments are carried out, and the effectiveness of each component is verified. It is worth noting that the geometric prior can only play a guiding role in the case of good decoupling. The feature visualization results show that the model has strong discrimination and is not sensitive to changes in the observation point.

Generally speaking, this study points out that in the process of cross-view person re-identification, the two types of information should be combined to achieve the purpose of robustness. The biggest advantage of this method is that it can unify the two technologies into a framework, which can provide a reliable path for the development of computer vision technology.

Keywords: Aerial-Ground Person Re-Identification, Geometric-Semantic Synergy, Cross-view Matching, Computer vision

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1. Introduction

The purpose of aerial ground person re-identification is to find the same person in the ground view and the aerial view, which is an important research content in the field of intelligent supervision. At present, the biggest problem is that the difference between viewpoints is relatively large, which leads to semantic ambiguity and geometric distortion^[1]. Traditional methods are based on the assumption that the viewpoints are consistent^[2], but AGPReID needs to deal with a large number of visual changes, and the traditional methods can not meet the requirements^[3].

Some methods, such as VDT and AG-ReID, can extract view-invariant features, but they also have three shortcomings. First, the geometric and semantic changes are too strong; second, the annotation cost is too high; third, the geometric alignment is not perfect.

In order to solve the above problems, this paper constructs a network framework, which is called GAP-Net, and the biggest advantage of this framework is that it can introduce adaptive semantic prompts into the geometric prior estimation. In this study, a lightweight convolutional module is introduced to extract explicit geometric prior information, and the prompt mechanism is self-calibrated to ensure that the prompt has strong geometric awareness, but the semantic invariance is not lost. The three large-scale datasets LAGPeR, G2APS-ReID and others are selected for a large number of experiments, and it is found that the performance of GAP-Net is relatively ideal.

1.1. Background and Motivation

Person re-identification is an important technology in intelligent monitoring, which can capture the same person through different cameras, and will not be affected by the background or lighting^[4]. With the development of UAV technology, this technology has been further upgraded, and aerial-person re-identification technology has been developed. Conventional ReID technology usually uses similar viewpoints to query and establish a database^[5], but AGPReID technology can capture people from different angles, such as ground cameras and drones^[6].

In recent years, the demand for ground and aerial monitoring systems has been increasing, especially the increasing number of drones, which can play an important role in disaster relief, traffic monitoring and public security. Think of a real-life scenario: the drone spots a person from the air, then the ground camera continues to track the person in the city-wide network. Or from the ground camera to find a person, and then use the drone to find the person in a large area. This is a two-way search, which makes AGPReID a powerful tool, not just an academic toy. If cross-view matching is not smooth, the surveillance system will be fragmented and the benefits of different types of cameras will not be brought into play.

In the process of solving AGPReID, we need to analyze why it is more difficult than the traditional one. We should make an assumption: the query image and the gallery image are taken from the same viewing angle. In this case, the changes in appearance are caused by occlusion, background clutter, light and other factors, and the body structure is relatively stable. The above methods have achieved good results in the above assumptions. The biggest problem with AGPReID is that it does not conform to this assumption. If the same person is observed from the ground and 50 meters above, the background and lighting are completely different, and the geometric structure has changed. Head-to-body ratio, limbs, body parts and so on have changed significantly. Therefore, the features extracted in the traditional way can not be used in the ground-air case.

In-depth analysis, the existence of geometric distortion is not an isolated problem, it is closely related to the occurrence of semantic changes, the two problems are interrelated, and the existence of one problem will lead to the other problem. The following is an example: if a person has a unique logo on his chest, the

logo can be seen from the ground, which can play an important role in identifying him, but from the air, because the camera is too high, the chest area is compressed and rotated, and the logo can not be seen, which is caused by geometric deformation, not because of the occlusion or the lighting is not good. In addition, when the aerial image is transformed into a ground image, the texture is missing, the interpolation artifact is generated, and the semantic quality is reduced. The above analysis further shows that semantic invariance and geometric alignment should be considered comprehensively.

Figure 1-1 describes how to deal with the challenges encountered in the process of forming such a view gap. Geometric distortion is the first challenge. Aerial views are generally steeper, which can easily lead to body deformation and scale changes, and some parts may be blocked. For example, the head occupies a large proportion of the aerial image, but the lower body is not visible at all^[7]. Semantic ambiguity is the second challenge. In addition to geometric changes, there are also changes in appearance, which are related to lighting, background and weather^[8]. The existence of these two factors makes it difficult to match identities, because the same person will have different appearances in the eyes of drones and fixed ground cameras.

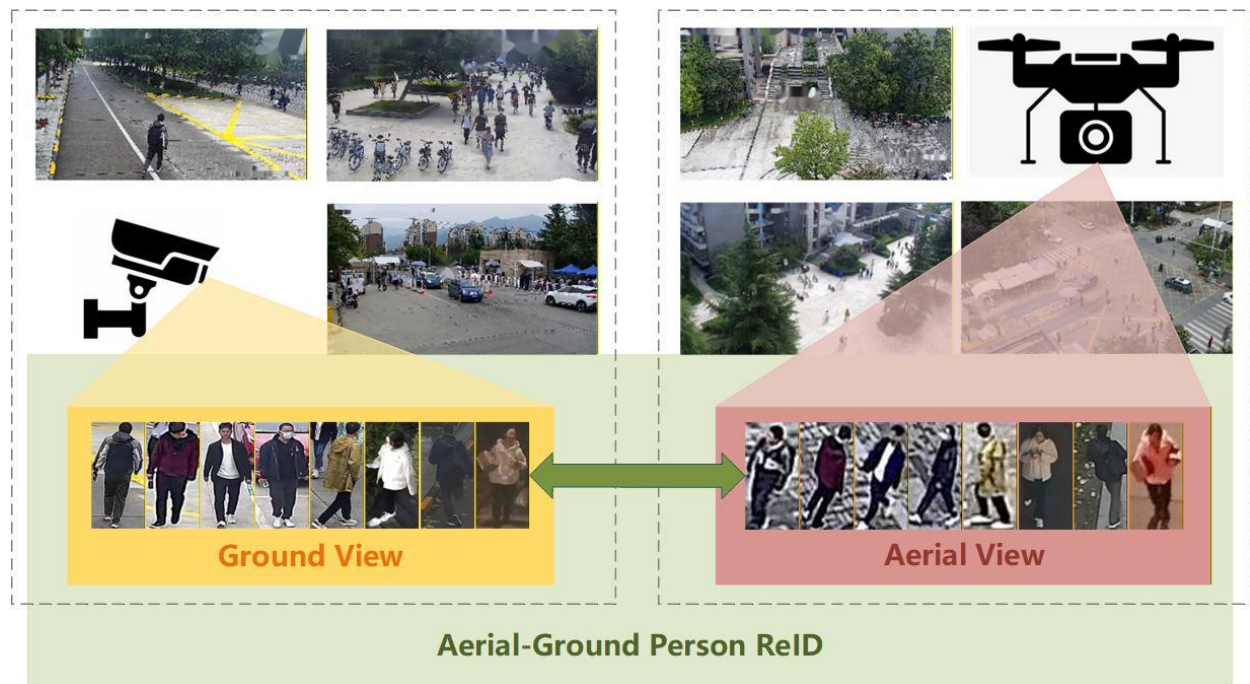


Figure 1-1 The significant appearance gap between aerial and ground views, caused by their extreme viewpoint differences, poses a substantial challenge for cross-view image matching.

What is the motivation for this research? The most critical observation is that AGPReID has strong coupling between semantic and geometric factors. If one factor is ignored and the other is emphasized, the feature representation is not ideal, which requires the construction of a unified framework to solve the problem.

1.2. Problem Statement

In the process of re-identification, it is necessary to ensure that the viewpoints of the two cameras are similar, and the cameras are installed at the same height, which is the premise of traditional re-identification^[9]. In the process of aerial and ground re-identification, the viewpoints are completely different, the aerial camera is installed on the drone, about 20-60 meters high, and the ground camera is fixed, which leads to the existence of huge gaps in the viewpoints. The first challenge is to make a reasonable geometric distortion, because the camera is installed at a higher position, the human body is

deformed, the scale is changed, and the second challenge is to deal with the ambiguity of semantics, because the color and texture of clothing and the illumination are different, the traditional re-identification technology can not be used, and the two factors are related, which can not be solved by simple alignment.

In the process of solving these problems, the progress achieved in recent years is relatively ideal. AG-ReID uses identity attributes to guide the model to identify invariant areas, which can reduce semantic ambiguity^[10]. VDT introduces view-decoupling transformer, which can separate view-related features from view-invariant ones, and can also weaken the impact of different viewpoints^[11]. Although these methods can get good results, there are still three problems that need to be solved urgently.

The most critical shortcoming is that there is no geometric alignment. Although the VDT has achieved good results in decoupling the information of the specific view, it has not made a reasonable model for the spatial transformation between different viewpoints. In the absence of geometric guidance, the decoupling characteristics are not strong enough to distinguish when the pose changes significantly. The second shortcoming is that the coupling between geometric and semantic changes has not been resolved. The two models pay too much attention to semantic invariance, but do not analyze the geometric distortion, which leads to the sensitivity of the characteristics they learn to spatial misalignment, especially when the body parts are deformed. The third shortcoming is that the annotation of attributes is too strong. In the process of training AG-ReID, the attribute label should be accurate, but in the real scenario, it is difficult to get such labels, and the cost is relatively high, which restricts the generalization and scalability of the method.

Generally speaking, the above limitations require the establishment of a unified framework to deal with semantic changes and geometric distortion, and do not need to pay a high cost for annotation, which can also guide the learning of features.

In the previous research, the two problems are not analyzed, one is the ambiguity of semantics, the other is the distortion of geometry, which are interrelated. It should be understood that the two factors are not independent of each other. If there is a change in the perspective, the human body will deform (this is the geometric problem), and the appearance will also change (this is the semantic problem). For example, if a person's backpack is observed from a high place, it is only a small patch and can not be identified, but if it is observed from the ground, it is a prominent feature. If you only solve the geometric alignment problem, you can only solve the semantic invariance problem, but you can not control the other factor. Therefore, we should build a reasonable framework and model the two factors at the same time, which is the main gap to be filled in this study.

1.3. Approach: GAP-Net

In order to solve the above problems, this paper introduces a new framework, GAP-Net, which can not only adapt to the semantic prompt, but also make up for the shortcomings of geometric prior estimation. This framework has three modules, each of which can play an important role in solving the problem.

The first is the GPE module, which makes up for the shortcomings of the previous geometric alignment. A lightweight convolutional subnetwork is designed to obtain the deformation field from the feature map, which can reflect how the human body changes in the case of different observation points. The biggest difference between this method and the previous geometric alignment method is that it does not need to twist the characteristics, but only to maintain the deformation field, so as to get a geometric prior and encode it into a compact geometric embedding E_{geo} , so as to provide effective geometric guidance for the model without increasing the number of feature streams, and the model has a strong adaptability to spatial deformation.

The second is the Geometric-Aware Prompt Calibration (G-PCM) module, which is used to solve the problem of coupling between semantics and geometry. This module builds on the prompt learning paradigm

from SeCap^[12], but the difference is that it can inject geometric embedding into the prompt calibration process. In this way, the geometric embedding E_{geo} is combined with the view-invariant features X_{inv} to obtain visual representation, and the visual representation can guide the calibration of the learnable prompt. The geometric-aware prompt P_{geo} can encode semantic invariance and geometric deformation pattern at the same time, which is exactly what we need to solve the problem of coupling.

Why do we use prompt learning as the basis for semantics? This is not a casual choice. The latest research results show that the use of learnable prompts can describe the semantics without the need for attribute annotations, and can achieve better annotation efficiency and stronger generalization ability. It should be noted that the spatial deformation is not considered in the process of prompt learning. In the process of prompt calibration, geometric prior knowledge can be introduced, and the model can find the right place and get the right information. The synergy between geometry and semantics is the most important contribution of this concept.

Third is the Local Feature Refinement Module (LFRM), which addresses the reliance on attribute annotations. Following SeCap, this module uses the geometric-aware prompts to refine local features through two-way attention mechanisms. Discriminative information is mined from local regions without needing any attribute annotations, and the refined features end up being robust to both geometric and semantic variations—so effective cross-view matching becomes possible.

When these three modules are put together, GAP-Net offers a holistic solution where geometric alignment and semantic invariance are integrated into a single, annotation-efficient framework. All of these components form a coherent pipeline in which geometric and semantic information mutually reinforce each other, allowing the model to learn robust view-invariant representations without having to rely on those costly attribute annotations.

1.4. Contributions

The main contributions of this work are summarized as follows:

- This study innovatively introduces a new method, which is the first framework to inject geometric priors into adaptive prompt learning, and the two challenges of semantic ambiguity and geometric distortion can be solved.
- A lightweight prior extraction module is constructed. In this module, the deformation field is estimated, which is compact and can be described by embedding without redundancy.
- In the experiment, the AGPReID data set was used to test the performance of the method, and it was found that the method had a strong advantage in the existing research results, and the two indicators were significantly improved. In addition, the effectiveness of each component is also verified.

2. Technical Background

Table 2-1 Statistical comparison of existing person re-identification datasets, encompassing view-homogeneous scenarios (solely ground or solely aerial) as well as view-heterogeneous aerial-ground scenarios.

Dataset	View	Data	#PersonID	#Camera	#Image	#Height
Market1501 ^[13]	Ground	Real	1,501	6	32,668	< 10m
DukeMTMC-reID ^[14]	Ground	Real	1,404	8	36,411	< 10m
PRAI1581 ^[15]	Aerial	Real	1,581	2	39,461	20~60m
UAVHuman ^[16]	Aerial	Real	1,144	1	41,290	2~8m
AG-ReID.v1	Aerial-Ground	Real	388	2(1A+1G)	21,893	15~45m
AG-ReID.v2 ^[17]	Aerial-Ground-Wearable	Real	1,615	3(1A+1G+1W)	100,502	15~45m
CARGO	Aerial-Ground	Synthetic	5,000	13(5A+8G)	108,563	5~75m
G2APS-ReID	Aerial-Ground	Real	2,788	2(1A+1G)	200,864	20~60m
LAGPeR	Aerial-Ground	Real	4,231	21(7A+14G)	63,841	20~60m

2.1. Single-View Person ReID under Homogeneous Camera Settings

Whether the camera angle changes, the model can accurately identify the same pedestrian, which is the key content of this study. Most of the traditional methods only focus on a single perspective, such as aerial view or ground view^[18]. In Table 2-1, it can be seen that the ground view has been studied more deeply, and a number of data sets have been created, including MSMT17^[19], Market1501 and so on. The development of these data sets has promoted the development of attention-based ReID and CNN-based ReID methods. In recent years, the performance of ReID has been significantly improved, which is enough to show that Vision Transformer (ViT)^[20] has strong feature extraction ability. TransReID has also achieved good results in the same period, because it has strong local feature modeling ability^[21]. In the process of aerial view ReID research, many problems need to be solved, such as occlusion, scale change, body deformation and so on. Some data sets have been built in this respect, including UAVHuman and so on. In methods such as the Rotation Invariant Transformer^[22] and Rotation Exploration Transformer^[23], attempts are made to reduce the impact of viewpoint changes through rotation-invariant modeling.

In the process of promoting the development of ReID, the prompt learning technology has achieved good results. The introduction of learnable prompt vector can make the model adapt to the downstream task, and can also retain the knowledge learned in the process of backbone training. In CLIP-ReID^[24] this technology is introduced into ReID for the first time, and Instruct-ReID^[25] further explores the prompt mechanism. SeCap uses PRM in the process of design, which is inspired by the above research, and can extract view-invariant characteristics through adaptive calibration.

Of course, the above methods can only deal with the same type of changes, such as changes in the ground camera, and can not deal with the changes caused by the difference between the ground and the air, which is not ideal.

In addition to CNN, the performance of Vision Transformer in ReID is also relatively ideal. The biggest difference between the two is that the latter can model the global context, while the former can only model the local context. In the process of processing, if the human body is deformed or the parts are not aligned, the latter can play an important role. In TransReID, side information embedding and patch shuffling are introduced to achieve good results in multiple benchmark tests. However, these methods can

only ensure the consistency of camera perspectives, but can not solve the problem of aerial-ground matching.

2.2. Cross-View Person ReID between Aerial and Ground Platforms

Aerial-Ground Cross-View Person Re-Identification is a new research direction, and the biggest challenge is to find the same person in the two pictures, one of which is taken from the drone and the other from the fixed camera. The biggest problem is that the shooting height, angle and lighting are different, and the same person will have a great difference in the two views, which will lead to semantic ambiguity and geometric distortion^[26].

2.2.1. Existing AGPreID Methods

AG-ReID series is a pioneering work in this field. AG-ReID.v1 first introduced the aerial-ground ReID task and used attribute labels to guide the model to invariant regions. AG-ReID.v2 has a larger data set scale and introduces EVA to make full use of the invariant information between the two views. However, the biggest problem of the two methods is that they rely too much on attribute annotation, which is not conducive to the generalization of the model.

A different approach is taken in VDT, where a ViT-based view decoupling mechanism is proposed to separate view-related features from view-invariant ones through hierarchical decoupling, and that does help reduce the impact of viewpoint differences. Significant improvements on datasets like CARGO are shown in experiments with VDT. But the main focus in VDT is on semantic-level decoupling, so its ability to handle geometric distortions is still pretty limited.

Then there is the Geometric and Semantic Alignment Network(GSAlign)^[27], where the geometric distortion problem is tackled more directly. A learnable Thin Plate Spline(LTPS)module and a Dynamic Alignment Module (DAM) are introduced. A set of keypoint displacements is learned in LTPS to adaptively warp pedestrian features, compensating for geometric distortions caused by viewpoint changes; visible region masks are estimated in DAM to reduce the effects of occlusion and partial observation.

In the process of prompt learning, a new solution is put forward. In this way, the prompt recalibration module can adjust the prompt according to the characteristics of the input, and the local feature refinement module can extract the invariant information from the local features.

Although the above methods have achieved certain results, the coupling relationship between geometric distortion and semantic ambiguity has not been solved.

2.2.2. Vision Transformer and Prompt Learning in ReID

ViT has become the backbone of the development of this technology, because it can use multi-header self-attention mechanism to deal with long-distance dependencies. CNNs are different in this regard. The receptive field is gradually increasing. ViT divides the input image into a series of patches, and the patches are processed in sequence, which can capture global information from the first layer. In the process of TransReID, overlapping patches and side information are introduced into ViT.

Prompt learning was born in NLP. The most fundamental idea is to introduce a small set of vectors, which are called prompts, and insert them into the input sequence of the transformer model. When training the model, these prompts can be used to complete the task, but most of the backbone is frozen, which can achieve the purpose of efficient adaptation. In the process of visual recognition, VPT has proved that this method can achieve the same effect as full fine-tuning, but the number of parameters is relatively small.

In the process of person matching, CLIP-ReID first used vision language for pre-training, and Instruct-ReID introduced prompt mechanism, which could play a role in multiple aspects. In this study, SeCap is the most relevant, which can dynamically adjust the prompt according to the input characteristics, and the prompt mechanism can be self-calibrated. Despite the fact that the above methods have been widely used, they are only semantic, can not model geometric deformation, and can not form a perfect mechanism.

2.2.3. Geometric Alignment in Cross-View Matching

In the process of geometric alignment, it is necessary to find the corresponding relationship between images from different perspectives. SIFT and other traditional technologies can estimate the displacement of pixels, but these technologies are only suitable for matching scenes, not for deformation at the level of human body. In the process of person reID, some scholars have adopted the method of part alignment, divided the human body into horizontal stripes, and can get good results if the body parts are in the same order, but if the viewing angle changes significantly, the hypothesis is no longer valid.

In recent years, TPS transformation has been widely used to describe non-rigid deformation. Compared with affine transformation, this method can not only capture non-linear warping, but also make the deformation smooth. GSAlign introduces the TPS module into AGPreID, which can be learned. However, this method also has some shortcomings, the most prominent of which is that the computational cost is relatively high, and the texture information near the occlusion boundary can not be described.

So if you look at where things stand, existing AGPreID methods are pretty fragmented. On one hand, you have semantic-oriented approaches like AG-ReID, VDT, and SeCap. On the other hand, there are geometric-oriented ones like GSAlign. Efficient semantic adaptation is provided by prompt learning, but no geometric awareness is present at all; spatial deformation is handled by geometric alignment, but it is not integrated with semantic learning. This fragmentation is exactly what motivates the proposed GAP-Net—geometric priors are explicitly injected into adaptive prompt learning, so both geometric distortions and semantic ambiguities can be handled together in one unified architecture.

2.3. Benchmark Datasets for Aerial-Ground Person Re-Identification

The scale and diversity of the data set are the key to the rapid development of AGPreID technology. AG-ReID.v1 is the first data set to complete this task, but its scale is relatively small. AG-ReID.v2 adds wearable devices to the data set, and although the scale has been improved, it still needs to rely on wearable devices to bridge the gap between ground and aerial, but this is not consistent with the real situation, because the data set does not contain wearable data.

CARGO and other synthetic data sets have the advantage of providing a large number of benchmarks with accurate viewpoints, but there are also problems, the most typical of which is the existence of domain gaps, which will lead to the decline of model performance in the real scene. The G2APS^[28] data set was reconstructed into G2APS-ReID to obtain a large amount of real-world data. The LAGPeR data set was independently constructed by the SeCap team, which has 4231 identities and 63841 images, and the images were captured with the help of 7 drones and 14 ground cameras in different real environments.

Combing the above research results, it is not difficult to see that the data set has achieved the transformation from small scale to large scale, from single model to multi model. In this study, two data sets are selected to evaluate the performance of GAP-Net, and the effectiveness of this method is verified.

3. The Problem

In the traditional ReID task, there are many challenges, the most critical of which is to find a reasonable way to deal with the difference between the two perspectives. Aerial camera height is generally 20-60 meters, and the shooting angle is relatively large, while the ground camera is relatively flat, and the perspective is completely different. This is a major challenge, which is reflected in two aspects: one is the ambiguity of semantics, the other is geometric distortion^[29].

3.1. Geometric Distortion

In the process of shooting, the angle changes too much, which leads to the deformation of the human body, and the human body is seriously compressed, resulting in occlusion. In the aerial view, the head occupies a large area of the image, while the legs are seriously compressed and even cropped. In the ground view, the human body is relatively normal, but there is a certain geometric deviation, which can not be matched at the pixel level.

In existing methods that try to handle geometric distortion—like GSAlign—reliance is placed on explicit spatial transformations such as learnable Thin Plate Spline warping to align features from different viewpoints. But several inherent limitations come with these approaches. First, a fair amount of computational overhead is added when the feature maps are warped, because interpolation and resampling operations are needed. Those operations don't just increase inference time; They can also cause information loss, especially near occlusion boundaries or regions with fine texture details. Second, the alignment process is usually kept separate from semantic feature learning. The features are warped first by the model, and then semantic cues are extracted from the warped representation, rather than both objectives being optimized together. This decoupling means the alignment cannot adapt to help semantic discrimination, and semantic cues cannot guide the alignment process either. Third, appearance ambiguities caused by lighting changes, occlusion, or differences in background cannot be fixed by alignment alone. Even when perfect geometric warping is done, the aerial and ground images can still look very different in color, illumination, and texture because they come from different camera sensors and environmental conditions. A more elegant solution is suggested by these limitations—one where geometric information is preserved without explicit warping and integrated smoothly with semantic learning.

3.2. Semantic Ambiguity

In the process of geometric changes, the external characteristics will change significantly, such as lighting, background, image preprocessing and so on. The differences between the two views are relatively large, which will lead to ambiguity in semantic information, and the color, texture and pattern of the same person's clothing will be different. Take a simple example: if someone wears a red coat, the color of the coat will be more saturated in the overcast ground environment, but if the same person is shot from the air, the color of the coat may be light, washed out, or even orange. In addition, the dark backpack will be completely consistent with the shadow of the person in the aerial view, but the ground view will be completely different from the background. In addition to additive noise, there are also many other factors that can change the visual characteristics of a person.

When the hardware of the two types of cameras is different, the image acquisition conditions are different, and the semantic ambiguity is more prominent. The camera used in the drone is lightweight, which can adjust the exposure automatically and has a strong white balance, but the lighting environment will change rapidly during the movement of the drone. The ground camera is fixed in position and will not move. Although the two cameras can capture the same person under the same lighting conditions, the color

temperature, contrast, noise level and other aspects are quite different, which will lead to a series of changes in the semantic characteristics and affect the matching of identity.

AG-ReID and other semantic methods can guide users to learn semantic characteristics that will not change because of changes in the perspective, and can also adjust the prompt, but they can not describe the geometric distortion. In the process of processing, although the appearance of the target will change, there is no effective way to deal with the geometric distortion. The problem is that even if the target has a strong robustness to the viewpoint, it will also change due to the spatial deformation, and the logo or head will not be identified^[30]. For example, the chest of the target has a logo, which can be identified from the ground, but can not be identified from the aerial view. In this case, the reason why semantic information is lost is not because of occlusion or lighting, but because of geometric deformation. This observation shows that if there is a serious geometric distortion, the semantic invariance can not be achieved.

3.3. The Coupled Nature of Geometric and Semantic Challenges

What should be paid attention to is that semantic ambiguity and geometric distortion are not independent factors, but have a close relationship. In order to understand how to achieve coupling, we need to analyze the specific examples. If a person has a unique mark on his chest, the mark can be identified from the ground, which is the key clue for matching. However, if we take aerial view, the chest area will be rotated and compressed, and the mark will not be identified, because the camera is in a vertical position. In this case, the semantic information is lost, which is inevitably related to geometric deformation.

Now, consider the opposite situation. Even if the aerial image is adjusted to a ground-like view, the warped area still has interpolation artifacts or lacks texture, which will reduce the semantic quality. The existence of two-way dependency means that the matching error caused by the gap between the two views is not only related to semantic error, but also to geometric error. A slight geometric misalignment combined with a moderate appearance change will lead to a sharp decline in retrieval accuracy, which can not be captured by evaluating the two factors separately. Therefore, we should find an effective solution to coordinate the relationship between semantic appearance and geometric deformation, and not regard them as independent problems.

The existence of this gap has prompted the need for a unified framework, which can bring geometric prior into the process of semantic feature learning, so that the model can deal with the above problems. The GAP-Net model can meet this requirement, which can not only estimate the geometric prior, but also calibrate the prompt adaptively, which will be introduced in detail in the next section.

4. The Solution

4.1. Overview

The overall framework of the Geometric-Aware Prompt Calibration Network (GAP-Net) is shown in Figure 4-1(a), which follows an enhanced encoder-decoder architecture.

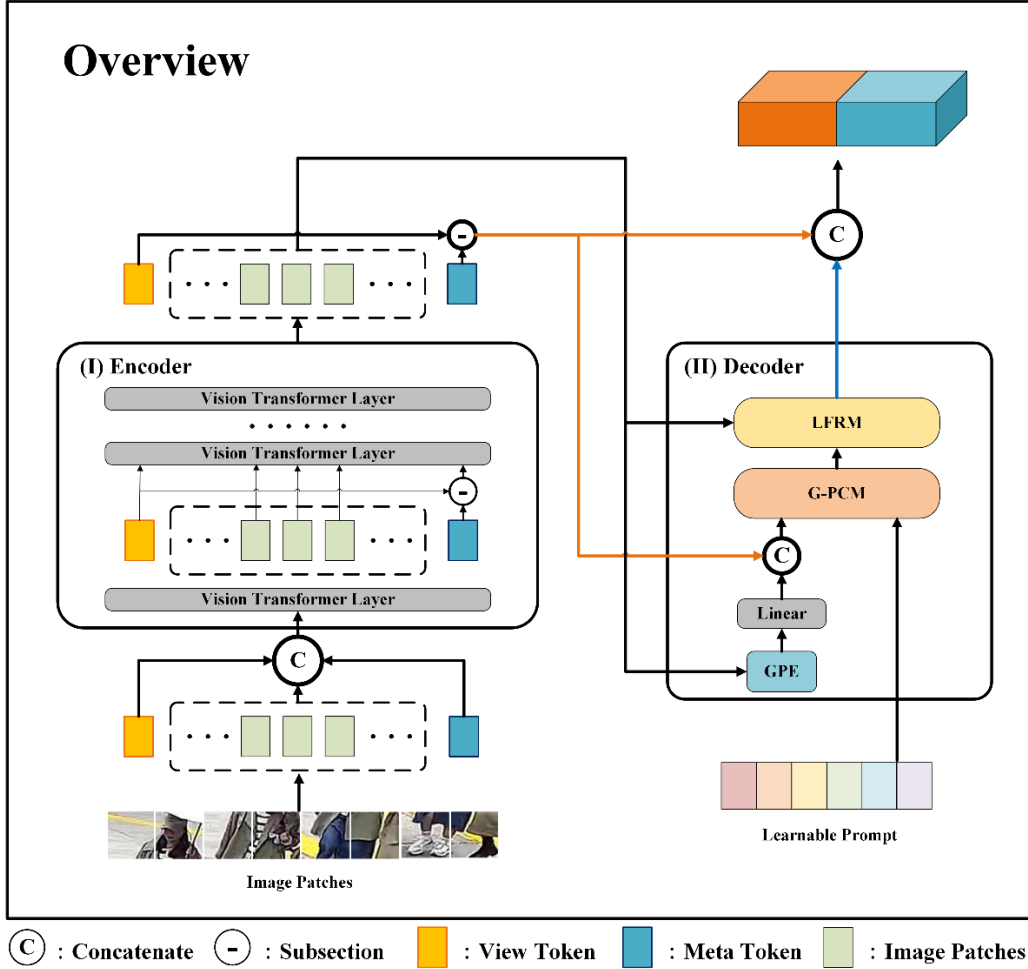


Figure 4-1 The architecture of the proposed GAP-Net

The encoder can extract visual features and process them in the process of extracting information, which will not be affected by the change of perspective. It is constructed on the basis of the view decoupling transformer, which can process the input image as a sequence of patches, and use the Meta token to process it. When processed by the vision transformer, the encoder can get two results: one is a viewpoint-invariant representation X_{inv} , and the other is a set of high-level local features.

The decoder can play a role in this case and play a role in the process of refining features, with three modules, namely G-PCM, LFRM and GPE. The following is an analysis of the role of the three modules. The encoder's feature map is used by GPE to estimate the deformation field, and the geometric embedding is obtained after distillation, which can describe the physical changes in the scene. Using this geometric embedding, the G-PCM module can be used to calibrate a series of learnable prompts to achieve semantic invariance and geometry awareness. At the same time, the LFRM module uses the geometric prompt to mine the information described by the local features with the help of bidirectional attention mechanism, and finally obtains the refined features.

The overall framework can be described as follows:

$$\begin{aligned}
 [X_{meta}, X_{view}, X_{local}] &= VDT([X_{meta}, X_{view}, [tokenization(X)]]) \\
 X_{inv} &= X_{meta} - X_{view} \\
 E_{geo} &= GPE(X_{local}) \\
 \tilde{X}_{inv} &= X_{inv} + Linear(E_{geo}) \\
 X_{local} &= LFRM(X_{local}, G - PCM(P^{(0)}, \tilde{X}_{inv})) \\
 Out &= [X_{inv}, X_{local}]
 \end{aligned} \tag{1}$$

where VDT is the View-Decoupling Transformer, X_{meta} and X_{view} are the Meta token and View token, $tokenization(X)$ refers to the process of converting the input image patches X into tokens, X_{inv} and $\tilde{X}_{inv}$ represents the view-invariant features, $P^{(0)}$ denotes initial learnable prompts, X_{local} signifies the local features of the input data, and Out is the final output.

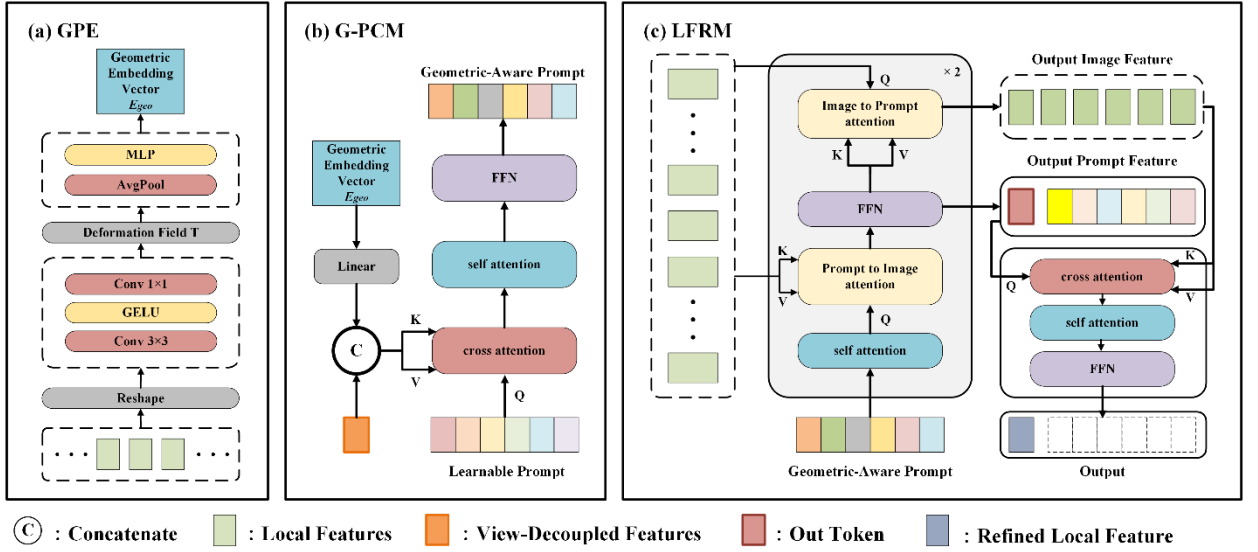


Figure 4-2 The framework of the (a) Geometric Prior Extraction (GPE) Module; (b) Geometric-Aware Prompt Calibration Module (G-PCM); (c) Local Feature Refinement Module (LFRM).

4.2. Geometric Prior Extraction Module

The GPE module can accurately grasp the spatial deformation pattern through the input image, and encode the pattern into a form of geometric information that can be used in subsequent modules. It is quite different from other methods. GPE can not only compress the deformation field into a low-dimensional space, but also predict the deformation field, so as to obtain more geometric information without changing the original feature space.

As can be seen from Figure 4-2 (a), this module initially reshaped the refined image patches from the encoder's last layer into the spatial feature map $F \in R^{H \times W \times C}$, then a lightweight convolutional network is employed to predict a dense deformation field $T \in R^{H \times W \times 2}$. The output of this sub-network is a two-channel map, where each spatial location $p = (x, y)$ corresponds to a 2D offset:

$$T(p) = (\Delta u(p), \Delta v(p)), \forall p \in \Omega \tag{2}$$

where Ω denotes the spatial grid of the feature map. Each offset $(\Delta u, \Delta v)$ indicates the estimated displacement of pixel p relative to a canonical viewpoint. This deformation field encapsulates the spatial distortion introduced by the aerial-ground viewpoint change.

In the process of prompt learning, deformation information should be integrated, and global average pooling T should be used first to obtain a certain size of deformation feature vector, and then geometric embedding $E_{geo} \in \mathbb{R}^d$ (where d is the prompt dimension) should be used to describe it, which needs to be mapped through two layers of MLP:

$$E_{geo} = MLP_{geo}(Flatten(AvgPool(T))) \quad (3)$$

4.3. Geometric-Aware Prompt Calibration Module

The G-PCM module is constructed with the help of Transformer Decoder, which is a geometric version. In this module, the geometric embedding is introduced to extract the characteristics that will not change with the change of the observation point, so that the prompt has strong adaptability and can play the role of geometric awareness. Specifically, the prompt is initialized with a set of vectors:

$$P^{(0)} = [p_1, p_2, \dots, p_L] \in \mathbb{R}^{L \times d} \quad (4)$$

where L is the hyperparameter that denotes the prompt length, and d is the prompt dimension, which means the initialized learnable prompts $P^{(0)}$ is a set containing L prompt vectors, each of which is d -dimensional.

As can be seen from Figure 4-2 (b), in order to ensure that the geometry embedding E_{geo} can be consistent with the dimension of the viewpoint-invariant features X_{inv} , it needs to be projected through a linear layer, and then the two are combined in the element level to form a new visual feature, which has a strong geometric enhancement:

$$\tilde{X}_{inv} = X_{inv} + Linear(E_{geo}) \quad (5)$$

In this process, the self-attention layer is introduced to realize the information exchange between the prompt sequence, and the global context cues are integrated into the prompt set. The calibration process is divided into three steps. First, cross-attention is carried out, and the prompt is regarded as the query (Q).

$$P^{(1)} = CrossAttention(P^{(0)}, \tilde{X}_{inv}, \tilde{X}_{inv}) \quad (6)$$

Next, the self-attention mechanism is used to process the prompt sequence, so that the information in the prompt can be exchanged, and the global context can be obtained:

$$P^{(2)} = SelfAttention(P^{(1)}) \quad (7)$$

The construction of the forward network is also a key content. It can make non-linear transformation, and the prompt obtained in this way can play a role in recalibration:

$$P_{geo} = FFN(P^{(2)}) \quad (8)$$

The resulting geometrically aware prompts P_{geo} encode both semantic view-invariance (from X_{inv}) and geometric deformation priors (from E_{geo}), providing a unified conditioning signal for the subsequent local feature refinement stage. The entire G-PCM process can be compactly expressed as:

$$P_{geo} = FFN(SA(CA(P^{(0)}, \tilde{X}_{inv}, \tilde{X}_{inv}))) + P^{(0)} \quad (9)$$

where SA represents self-attention mechanism, CA refers to the cross-attention mechanism.

4.4. Local Feature Refinement Module

The purpose of constructing LFRM is to obtain view-invariant information from local features. The geometrically-aware prompts can play a guiding role in this process. The structure of this module is consistent with that of SeCap, but the input prompts are replaced with geometric-aware prompts P_{geo} , as shown in Figure 4-2 (c).

LFRM uses the decoder structure and has two modules, one is the two-way attention module, the other is the feature fusion module. Given the geometrically aware prompts P_{geo} and the local image features F_{local} , the two-way attention module enables both prompt-to-image encoding and image-to-prompt encoding directions. In the process of attention from prompt to image, the prompt P_{geo} is regarded as the query to obtain the corresponding information from the local features. Before the cross-attention, the self-attention layer is used to process the prompt, so as to grasp the internal relationship between them. After the prompt is updated, it is input into the cross-attention layer, and then the local features are paid attention to, and the FFN is used to connect the residual, as shown in the following figure:

$$F_p = FFN(CA(SA(P_{geo}, F_{local}, F_{local}))) + P_{geo} \quad (10)$$

here, F_p represents refined prompt features that have been enriched with image-specific information, FFN stands for feed-forward network, CA represents cross-attention mechanism and SA represents self-attention mechanism, P_{geo} is the geometrically aware prompts and finally F_{local} signifies the local image features.

Second, the role of image-to-prompt attention is to reverse: the local characteristics can be regarded as query, which can pay attention to the refined prompt features F_p . In this step, each local patch can incorporate cues from the prompt, including geometry and semantics. The operation is as follows:

$$F_I = CA(F_{local}, F_p, F_p)) + F_{local} \quad (11)$$

where F_I denotes the image features enhanced by the prompt information. The two-way attention mechanism ensures that prompts become more image-adaptive while local features simultaneously become more semantically informed, creating a synergistic refinement process.

In order to make the decoder more lightweight, only two two-way attention blocks are set in LFRM. After the two iterations of two-way attention, the image features F_I are fused with the prompt features F_p to form the output local feature representation, which is processed by the feature fusion module. The cross attention layer uses the prompt features as the query and the image features as the key and value, so that the final representation can integrate the information of the two modalities. The self-attention layer and the forward network can refine the representation, as shown in the following figure:

$$[\mathbf{Out}, _] = FFN(SA(CA([\mathbf{Out}, F_p], F_I, F_I))) \quad (12)$$

where $\mathbf{Out}$ serves as the output token, which integrated final locally refined feature, F_p is the refined prompt features and F_I denotes the image features.

4.5. Optimization

GAP-Net uses a multi-task loss function to train the network in an end-to-end manner, which can optimize all the components in the framework. The overall loss is composed of five parts: (1) Identity classification loss, (2) Triplet loss, (3) View classification loss, (4) Orthogonality Loss and (5) Geometric regularization loss.

Identity Classification Loss. Following standard practice in person re-identification, the identity classification loss is applied to both global and local features, and the cross-entropy loss is used to train a classifier that predicts identity label of each input image, the identity classification losses are defined as:

$$\begin{aligned} \mathcal{L}_{ID}^{global} &= - \sum_{i=1}^N y_i \log(\hat{y}_i^{global}) \\ \mathcal{L}_{ID}^{local} &= - \sum_{i=1}^N y_i \log(\hat{y}_i^{local}) \end{aligned} \quad (13)$$

where N is the batch size, y_i is the ground-truth identity label of the i -th sample, and $\hat{y}_i^{global}$ and $\hat{y}_i^{local}$ are the predicted identity probabilities from the global and local features respectively.

Triplet Loss. In the process of completing the task, this loss function is also used. The purpose of using this function is to make the distance between the anchor and the positive sample as small as possible, and to make the distance between the anchor and the negative sample as large as possible. This function can be used in both global and local features, and the definition is as follows:

$$\begin{aligned}\mathcal{L}_{Tri}^{global} &= \frac{1}{N} \sum_{i=1}^N \max(0, d_{ap}^{global} - d_{an}^{global} + m) \\ \mathcal{L}_{Tri}^{local} &= \frac{1}{N} \sum_{i=1}^N \max(0, d_{ap}^{local} - d_{an}^{local} + m)\end{aligned}\quad (14)$$

where d_{ap} and d_{an} denote the Euclidean distances between anchor-positive and anchor-negative pairs, respectively, and m is the margin hyperparameter.

View Classification Loss. In order to obtain the view-related information, the view classification loss is introduced to supervise the VDT encoder to extract the corresponding features, which is defined as:

$$\mathcal{L}_{view} = - \sum_{i=1}^N z_i \log(\hat{z}_i) \quad (15)$$

where z_i is the ground-truth view label (aerial or ground) of the i -th sample, $\hat{z}_i$ is the predicted view probability from the view-related feature, and N is the total number of samples in the dataset.

Orthogonality Loss. In order to achieve the goal of complete decoupling, orthogonality loss is introduced to reduce the correlation between the two feature representations. The formula is as follows:

$$\mathcal{L}_{orth} = \sum_{i=1}^d |\langle \mathbf{inv}_i, \mathbf{v}_i \rangle| \quad (16)$$

where $|\langle \cdot, \cdot \rangle|$ denotes the absolute value of the dot product between two features, d is the feature dimension, and $\mathbf{inv}_i$ and $\mathbf{v}_i$ represent the i -th dimension of the view-invariant and view-related features, respectively.

Geometric Regularization Loss. In order to avoid the occurrence of irregular deformation field in the GPE module, this paper introduces geometric regularization loss, which can play the role of smoothness prior, so that the deformation field T is more reasonable and conforms to physical laws. The geometric regularization loss is defined as:

$$\mathcal{L}_{geo} = ||T^2|| = \sqrt{\sum_{p \in \Omega} ((\Delta u(p))^2 + (\Delta v(p))^2)} \quad (17)$$

where T is the dense deformation field, Ω denotes the spatial grid of the feature map, $\Delta u(p)$ and $\Delta v(p)$ are the predicted offsets at pixel location p .

Overall Loss. The above content has explained that in the process of training GAP-Net, the loss function can play an important role, and the above components are integrated into the loss function:

$$\mathcal{L} = \alpha(\mathcal{L}_{ID}^{global} + \mathcal{L}_{Tri}^{global}) + \beta(\mathcal{L}_{ID}^{local} + \mathcal{L}_{Tri}^{local}) + \lambda(\mathcal{L}_{view} + \mathcal{L}_{orth}) + \gamma\mathcal{L}_{geo} \quad (18)$$

where α , β , λ , and γ are hyperparameters that balance the contributions of different loss components.

5. Evaluation

5.1. Experimental Settings

5.1.1. Datasets

The two AGPreID data sets are used to evaluate the performance of the algorithm, namely G2APS-ReID and LAGPeR. Table 2-1 gives the statistical data of the two data sets.

LAGPeR is a data set built by the SeCap team to meet the needs of real-world AGPreID, with 63841 images, 4231 IDs, and 21 cameras, including 7 drones and 14 ground cameras. 2708 IDs are used to train the model, and the remaining 1523 IDs are used to test the model. For evaluation, three experimental settings are defined: $A \rightarrow G$ (aerial query with ground gallery), $G \rightarrow A$ (ground query with aerial gallery), and $G \rightarrow A + G$ (ground query with combined aerial and ground gallery).

G2APS-ReID dataset is reconstructed from G2APS dataset, including 200864 images, 2788 identities, and two cameras, one aerial and one ground. The training set includes 1509 identities and the test set includes 1219 identities. Two experimental settings are employed: $A \rightarrow G$ and $G \rightarrow A$. Since each identity has only one ground camera view, the $G \rightarrow A + G$ setting is not applicable.

5.1.2. Metrics

In the process of reidentifying people, the retrieval performance of GAP-Net is analysed from two aspects: (1) Rank-1 (2) Mean Average Precision (mAP).

In the process of calculating Rank- k accuracy, the proportion of query samples is taken as the basis, and the proportion of correct identity in the retrieved results is analysed. This study focuses on the analysis of Rank1, which is the most accurate retrieval result. mAP is a comprehensive evaluation, which can describe the ranking quality of all retrieved results. For each query, the average precision value is calculated to obtain the average precision (AP). The value of mAP is the average of AP.

The two indicators can reflect the retrieval performance from different aspects: the former can find the matching point, and the latter can reflect the ranking quality. The results of the above research do not use post-processing technology, which is compared with the previous research.

5.1.3. Implementation Details

All experiments are carried out using the PyTorch library and running on a single NVIDIA RTX 4090 GPU. The backbone network of the proposed GAP-Net adopts the ViT pre-trained on ImageNet^[31]. During both training and inference, input images are uniformly resized to 256×128 pixels. In the tokenization stage, the patch size and stride are both set to 16×16 . Following the standard ViT design, the embedding dimension d for each token is 768.

To enhance generalization, we apply common data augmentation strategies during training, namely random cropping, color jittering, and random erasing. The batch size of 64 is used, which comprises 16 distinct identities, each contributing 4 images. For triplet loss, the soft margin variant is adopted to avoid manual tuning of the margin hyperparameter m . The model undergoes training for 120 epochs, optimized by Stochastic Gradient Descent (SGD) with momentum. A cosine annealing learning rate scheduler is employed, starting from 8×10^{-3} and gradually decaying to 1.6×10^{-6} . During inference, no data augmentation or re-ranking techniques were applied, so as to ensure a fair comparison with existing methods.

5.2. Performance of GAP-Net

To comprehensively evaluate the proposed GAP-Net, comparisons are made against several state-of-the-art methods on the LAGPeR and G2APS-ReID datasets. For a fair comparison, the focus is placed exclusively on Transformer-based architectures, particularly those using ViT as the backbone. The compared methods include single-view ReID methods like ViT, TransReID, and CLIP-ReID; AGPreID methods like AG-ReID, VDT, and SeCap; and finally a cross-modal ReID method, MIP, which was re-implemented for the AGPreID task. All results are reported under the same evaluation protocols, without re-ranking or post-processing techniques.

As shown in Table 5-1, the $A \rightarrow G$ denotes that the Aerial view is the query, $G \rightarrow A$ denotes that the Ground view is the query, and $G \rightarrow A + G$ indicates that the gallery contains images from both the Aerial and Ground view. CLIP – ReID* indicates using OLP (Overlapping Patches) and SIE (Side Information Embeddings) in Clip-ReID. MIP[†]^[32] represents the re-implementation for the AGPreID. AG-ReID[‡] indicates removing the attributes branch of the AG-ReID method. The best performance is in **bold**.

Table 5-1 Comparison of the performance of outstanding methods on LAGPeR and G2APS-ReID datasets.

Method	BackBone	LAGPeR						G2APS-ReID			
		$A \rightarrow G$		$G \rightarrow A$		$G \rightarrow A + G$		$A \rightarrow G$		$G \rightarrow A$	
		Rank-1	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP
ViT	ViT	38.67	27.25	32.04	30.69	18.88	15.31	69.38	52.17	67.16	52.22
TransReID	ViT	38.80	28.80	33.00	32.10	22.90	18.80	67.10	53.10	68.52	54.19
CLIP-ReID	CLIP	24.40	17.60	21.30	20.80	12.30	10.20	58.30	42.20	56.41	41.92
CLIP-ReID*	CLIP	23.10	17.50	20.00	20.30	9.00	8.40	59.60	42.70	56.39	42.52
MIP [†]	ViT	39.30	29.30	33.90	32.60	21.00	17.30	73.00	57.40	70.22	57.06
AG-ReID [‡]	ViT	40.48	28.89	32.96	31.91	22.03	17.89	70.75	52.87	68.70	53.39
VDT	ViT	40.15	28.97	33.55	31.98	19.50	16.45	73.05	56.23	71.08	56.01
SeCap	ViT	41.43	29.76	33.55	32.03	22.36	18.07	73.91	57.94	69.26	56.54
GAP-Net	ViT	40.94	29.82	33.75	32.15	22.06	19.02	74.42	58.34	69.13	56.71

Generally speaking, the model constructed in this study achieves **state-of-the-art (SOTA)** in the two data sets, and the performance is relatively high in several settings. Compared with the baseline model, the advantages of this model are reflected in all evaluation settings, which fully proves that the framework constructed in this study is reasonable and effective. In the LAGPeR data set, the model has a high accuracy of 40.94% and a mAP of 29.82% in the $A \rightarrow G$ setting, and the accuracy reaches 33.75% in the $G \rightarrow A$ setting, with a mAP of 32.15%. In the G2APS-ReID data set, the model has a mAP of 57.34% and a high accuracy of 73.42% in the $A \rightarrow G$ setting, which is enough to show that the model has a strong ability to deal with cross-view matching problems.

Compared with the single-view method, the performance of GAP-Net is significantly higher than that of CLIP-ReID, TransReID and other methods in all cross-view settings. For example, in the LAGPeR dataset, the baseline of ViT is 2.27% in Rank-1 and 2.57% in mAP for the $A \rightarrow G$ setting. In the $G \rightarrow A$ setting, the baseline of ViT is 1.71% in Rank-1 and 1.46% in mAP. In the G2APS-ReID dataset, the improvement is more obvious. In the $A \rightarrow G$ setting, the baseline of ViT is 4.04% in Rank-1 and 5.17% in mAP. It can be seen that the introduction of geometric prior into the prompt learning can effectively solve the problem of visual bias, which is the key to solve the problem of single-view method.

When compared to existing AGPreID methods, GAP-Net shows competitive performance. The superior performance of GAP-Net can be attributed to its synergistic integration of geometric and semantic information. Geometric priors are explicitly injected into the prompt calibration process in GAP-Net, which allows the model to account for spatial deformations while view-invariant features are being extracted. On the other hand, geometric cues are leveraged in GAP-Net to guide the attention mechanisms within the LFRM, which results in more robust local feature refinement. The consistent improvements observed across both datasets validate that geometric-semantic synergy is essential for handling the coupled challenges inherent in AGPreID.

5.3. Ablation Study

In order to evaluate the role of each module in this method, this paper conducts ablation experiments, as shown in the table below. In the table, "Baseline" means that ViT is used as the backbone network in the ReID method, "LFRM" means that the local features are refined, "VDT" means that the view is decoupled, and "PRM" means that the calibration module is set. The best performance and the most prominent improvements are in **bold**.

Table 5-2 Ablation study of the components in GAP-Net evaluated on the LAGPeR dataset.

No.	Method	LAGPeR					
		$A \rightarrow G$		$G \rightarrow A$		$G \rightarrow A + G$	
		Rank-1	mAP	Rank-1	mAP	Rank-1	mAP
1	Baseline (ViT)	38.67	27.25	32.04	30.69	18.88	15.31
2	+VDT	40.15	28.97	33.55	31.98	19.50	16.45
		+1.48	+1.72	+1.51	+1.29	+0.62	+1.14
3	+LFRM	40.48	29.17	33.59	31.90	22.49	18.20
		+1.81	+1.92	+1.55	+1.21	+3.61	+2.89
4	+LFRM+G-PCM	40.51	29.19	34.11	32.05	22.98	18.20
		+1.84	+1.94	+2.07	+1.36	+4.1	+2.89
5	+LFRM+G-PCM+GPE	38.74	28.32	32.67	31.29	21.41	17.49
		+0.07	+1.07	+0.63	+0.6	+2.53	+2.18
6	+LFRM+VDT	39.43	27.96	31.32	30.13	19.57	15.70
		+0.76	+0.71	-0.72	-0.56	+0.69	+0.39
7	+VDT+LFRM+G-PCM+GPE (Proposed)	40.94	29.82	33.75	32.15	22.06	18.20
		+2.27	+2.57	+1.71	+1.46	+3.18	+2.89

In the comparison between #1 and #2, it is found that the integration of VDT can play a role in all aspects, which is enough to show that the model has a strong ability to resist interference through decoupling. In the comparison between #2 and #3, it is found that the introduction of LFRM can bring performance improvement, which means that the model can use this module to refine local features and achieve the purpose of cross-view matching.

In the above two cases, the performance of #3 is significantly improved by the introduction of G-PCM, while the performance of #5 is relatively poor compared with that of #6, which is related to the lack of VDT, resulting in the introduction of noise priors, which fully shows that the introduction of GPE can provide guidance in geometry.

The full model is GAP-Net (#7), which integrates VDT, G-PCM and GPE to achieve the best performance in this configuration: (1) VDT can decouple the viewpoint information; (2) GPE can obtain the information provided by VDT and play a role; (3) G-PCM can play a complementary role; (4) LFRM can refine the local characteristics. Compared with the baseline, the increase of 2.27% in Rank-1 and 2.57%

in mAP in $A \rightarrow G$, and by 3.18% and 2.89% in $G \rightarrow A + G$ is enough to show that the full model can play a role in AGPReID and enhance the synergy between geometry and semantics.

5.4. Feature Visualization

In order to evaluate the performance of the algorithm in learning cross-view features, this study takes tSNE as the tool to visualize the features of the two models on the LAGPeR dataset. The extracted features are projected into a 2D space.

From the figure, we can see that the circles represent the aerial view, the pluses represent the ground view, and the same color represents the same ID. It can be seen that the feature embedding obtained by the network is more perfect, and the same identity can be clustered, which has strong intraclass cohesion. The separation between different identities is more obvious, which shows that the interclass discrimination ability is stronger. Notably, the features of the same identity are well aligned in the embedding space, which fully shows that the network can learn view-invariant representations.

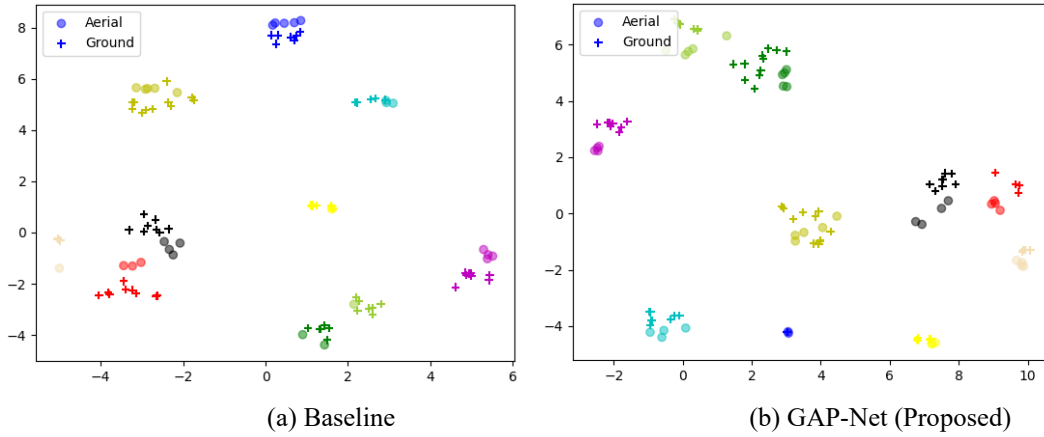


Figure 5-1 t-SNE visualization of feature embeddings extracted by the baseline model and the proposed GAP-Net..

The reason why the discriminability of features has been significantly improved is that the geometric-semantic design of GAP-Net is more reasonable. GPE module can provide good geometric clues for the model, guide the model to focus on the deformation area, and inject geometric priors into the process of prompt learning through G-PCM module. When combined with LFRM, the model can extract robust features.

6. Conclusions

AGPReID is a technology that can identify the identity of the same person from different perspectives, which can make up for the shortcomings of fixed cameras and has strong practicability. In this study, a framework is built to process geometry and obtain semantic information.

The geometric information and semantic information are closely related. In the previous research, the semantic information and geometric information are processed in different ways, and the two are analyzed separately. In this study, three ingredients are fully integrated. First, the geometric prior is extracted from the feature map, and the geometric information is encoded through the deformation method, but the features themselves will not change. Secondly, geometric prior is introduced into prompt learning, and the third is to refine the features through bidirectional attention, and the geometric prior is introduced into the prompt. In this way, the model can not only identify the geometric invariance, but also have a strong discrimination ability.

In this way, the image features are obtained, which have a strong robustness to the changes in the view, and we believe that this is the main reason for the good performance of GAP-Net in a variety of data sets (the results of the three data sets are similar). Ablation experiments further confirm that each component can bring a good contribution, and the introduction of geometric prior can play a key role in eliminating the confusion between different views. The results obtained in this study show that the coupling of semantics and geometry can produce significant advantages in the training process, and it is a reasonable view that the model can learn geospatial information.

In the process of aerial ground person re identification, we should not only pay attention to the change of view, but also take into account the impact of weather, lighting and other factors, and the change of scene type will also lead to adverse effects. In the next stage of research, we should make efforts to obtain robust person identity information and not be disturbed by the above factors. In the next stage of research, we should make efforts to obtain robust person identity information and not be disturbed by the above factors. In the next stage of research, we should make efforts to obtain robust person identity information and not be disturbed by the above factors.

Generally speaking, the introduction of geometric prior can play an important role in the process of semantic prompt learning, and can play a guiding role in the process of identifying people from different perspectives. Combining semantic invariance with geometric alignment can provide a new direction for solving the problem of extreme view change, and can also play an important role in solving other computer vision problems. It is expected that this study will lay the foundation for the next stage of research on how to interact between semantic and geometric representation in cross-view scenarios.

References

- [1] G. Lee, S. Lee, D. Kim, Y. Shin, Y. Yoon, and B. Ham, ‘Camera-Driven Representation Learning for Unsupervised Domain Adaptive Person Re-identification’, in *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, Oct. 2023, pp. 11419–11428. doi: 10.1109/ICCV51070.2023.01052.
- [2] Q. Zhang, J.-H. Lai, Z. Feng, and X. Xie, ‘Uncertainty Modeling with Second-Order Transformer for Group Re-identification’, *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 3, pp. 3318–3325, Jun. 2022, doi: 10.1609/aaai.v36i3.20241.
- [3] H. Li, M. Ye, and B. Du, ‘WePerson: Learning a Generalized Re-identification Model from All-weather Virtual Data’, in *Proceedings of the 29th ACM International Conference on Multimedia*, in MM ’21. New York, NY, USA: Association for Computing Machinery, Oct. 2021, pp. 3115–3123. doi: 10.1145/3474085.3475455.
- [4] X. Qian *et al.*, ‘Pose-Normalized Image Generation for Person Re-identification’, Apr. 25, 2018, *arXiv: arXiv:1712.02225*. doi: 10.48550/arXiv.1712.02225.
- [5] T. Chen *et al.*, ‘ABD-Net: Attentive but Diverse Person Re-Identification’, in *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, Oct. 2019, pp. 8350–8360. doi: 10.1109/ICCV.2019.00844.
- [6] W. Li *et al.*, ‘DC-Former: Diverse and Compact Transformer for Person Re-Identification’, Feb. 28, 2023, *arXiv: arXiv:2302.14335*. doi: 10.48550/arXiv.2302.14335.
- [7] L. Zheng, L. Shen, L. Tian, S. Wang, J. Wang, and Q. Tian, ‘Scalable Person Re-identification: A Benchmark’, in *2015 IEEE International Conference on Computer Vision (ICCV)*, Dec. 2015, pp. 1116–1124. doi: 10.1109/ICCV.2015.133.
- [8] J. Miao, Y. Wu, P. Liu, Y. Ding, and Y. Yang, ‘Pose-Guided Feature Alignment for Occluded Person Re-Identification’, in *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, Oct. 2019, pp. 542–551. doi: 10.1109/ICCV.2019.00063.
- [9] H. Luo, Y. Gu, X. Liao, S. Lai, and W. Jiang, ‘Bag of Tricks and a Strong Baseline for Deep Person Re-Identification’, in *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, Jun. 2019, pp. 1487–1495. doi: 10.1109/CVPRW.2019.00190.
- [10] H. Nguyen, K. Nguyen, S. Sridharan, and C. Fookes, ‘Aerial-Ground Person Re-ID’, Aug. 14, 2023, *arXiv: arXiv:2303.08597*. doi: 10.48550/arXiv.2303.08597.
- [11] Q. Zhang, L. Wang, V. M. Patel, X. Xie, and J. Lai, ‘View-decoupled Transformer for Person Re-identification under Aerial-ground Camera Network’, Mar. 21, 2024, *arXiv: arXiv:2403.14513*. doi: 10.48550/arXiv.2403.14513.
- [12] S. Wang, Y. Wang, R. Wu, B. Jiao, W. Wang, and P. Wang, ‘SeCap: Self-Calibrating and Adaptive Prompts for Cross-view Person Re-Identification in Aerial-Ground Networks’, Apr. 10, 2025, *arXiv: arXiv:2503.06965*. doi: 10.48550/arXiv.2503.06965.
- [13] L. Zheng, L. Shen, L. Tian, S. Wang, J. Bu, and Q. Tian, ‘Person Re-identification Meets Image Search’, Feb. 07, 2015, *arXiv: arXiv:1502.02171*. doi: 10.48550/arXiv.1502.02171.
- [14] Z. Zheng, L. Zheng, and Y. Yang, ‘Unlabeled Samples Generated by GAN Improve the Person Re-identification Baseline in Vitro’, in *2017 IEEE International Conference on Computer Vision (ICCV)*, Venice: IEEE, Oct. 2017, pp. 3774–3782. doi: 10.1109/ICCV.2017.405.
- [15] S. Zhang *et al.*, ‘Person Re-identification in Aerial Imagery’, *IEEE Trans. Multimedia*, vol. 23, pp. 281–291, 2021, doi: 10.1109/TMM.2020.2977528.
- [16] T. Li, J. Liu, W. Zhang, Y. Ni, W. Wang, and Z. Li, ‘UAV-Human: A Large Benchmark for Human Behavior Understanding with Unmanned Aerial Vehicles’, in *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Nashville, TN, USA: IEEE, Jun. 2021, pp. 16261–16270. doi: 10.1109/CVPR46437.2021.01600.
- [17] H. Nguyen, K. Nguyen, S. Sridharan, and C. Fookes, ‘AG-ReID.v2: Bridging Aerial and Ground Views for Person Re-identification’, Apr. 07, 2024, *arXiv: arXiv:2401.02634*. doi: 10.48550/arXiv.2401.02634.

- [18] Y. Qin, Y. Chen, D. Peng, X. Peng, J. T. Zhou, and P. Hu, ‘Noisy-Correspondence Learning for Text-to-Image Person Re-identification’, Mar. 28, 2024, *arXiv*: arXiv:2308.09911. doi: 10.48550/arXiv.2308.09911.
- [19] L. Wei, S. Zhang, W. Gao, and Q. Tian, ‘Person Transfer GAN to Bridge Domain Gap for Person Re-identification’, in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Salt Lake City, UT, USA: IEEE, Jun. 2018, pp. 79–88. doi: 10.1109/CVPR.2018.00016.
- [20] Z. Liu *et al.*, ‘Swin Transformer: Hierarchical Vision Transformer using Shifted Windows’, Aug. 17, 2021, *arXiv*: arXiv:2103.14030. doi: 10.48550/arXiv.2103.14030.
- [21] S. He, H. Luo, P. Wang, F. Wang, H. Li, and W. Jiang, ‘TransReID: Transformer-Based Object Re-Identification’.
- [22] S. Chen, M. Ye, and B. Du, ‘Rotation Invariant Transformer for Recognizing Object in UAVs’, Nov. 05, 2023, *arXiv*: arXiv:2311.02559. doi: 10.48550/arXiv.2311.02559.
- [23] L. Wang, Q. Zhang, J. Qiu, and J. Lai, ‘Rotation Exploration Transformer for Aerial Person Re-identification’, in *2024 IEEE International Conference on Multimedia and Expo (ICME)*, Jul. 2024, pp. 1–6. doi: 10.1109/ICME57554.2024.10687588.
- [24] S. Li, L. Sun, and Q. Li, ‘CLIP-ReID: Exploiting Vision-Language Model for Image Re-Identification without Concrete Text Labels’, Jan. 01, 2023, *arXiv*: arXiv:2211.13977. doi: 10.48550/arXiv.2211.13977.
- [25] W. He *et al.*, ‘Instruct-ReID: A Multi-purpose Person Re-identification Task with Instructions’, Apr. 29, 2025, *arXiv*: arXiv:2306.07520. doi: 10.48550/arXiv.2306.07520.
- [26] Q. Zhang, K. Dang, J.-H. Lai, Z. Feng, and X. Xie, ‘Modeling 3D Layout For Group Re-Identification’, in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2022, pp. 7502–7510. doi: 10.1109/CVPR52688.2022.00736.
- [27] Q. Li, J. Li, Y. Zhang, L. Tan, J. Chen, and J. Ji, ‘GSAlign: Geometric and Semantic Alignment Network for Aerial-Ground Person Re-Identification’, Jan. 13, 2026, *arXiv*: arXiv:2510.22268. doi: 10.48550/arXiv.2510.22268.
- [28] S. Zhang *et al.*, ‘Ground-to-Aerial Person Search: Benchmark Dataset and Approach’, in *Proceedings of the 31st ACM International Conference on Multimedia*, Oct. 2023, pp. 789–799. doi: 10.1145/3581783.3612105.
- [29] E. Ning, Y. Wang, C. Wang, H. Zhang, and X. Ning, ‘Enhancement, integration, expansion: Activating representation of detailed features for occluded person re-identification’, *Neural Networks*, vol. 169, pp. 532–541, Jan. 2024, doi: 10.1016/j.neunet.2023.11.003.
- [30] Y. Sun, L. Zheng, Y. Li, Y. Yang, Q. Tian, and S. Wang, ‘Learning Part-based Convolutional Features for Person Re-Identification’, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 43, no. 3, pp. 902–917, Mar. 2021, doi: 10.1109/TPAMI.2019.2938523.
- [31] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, ‘ImageNet: A large-scale hierarchical image database’, in *2009 IEEE Conference on Computer Vision and Pattern Recognition*, Jun. 2009, pp. 248–255. doi: 10.1109/CVPR.2009.5206848.
- [32] R. Wu, B. Jiao, M. Liu, S. Wang, W. Wang, and P. Wang, ‘Enhancing Visible-Infrared Person Re-Identification With Modality- and Instance-Aware Adaptation Learning’, *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 35, no. 8, pp. 8086–8103, Aug. 2025, doi: 10.1109/TCSVT.2025.3560118.

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